Classification of ADHD and Control Subjects Using EEG Theta Band Power and Machine Learning Approaches

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***Abstract*—Attention-Deficit/ Hyperactive disorder (ADHD) is a common neurodevelopmental condition. It is usually diagnosed through behavioral assessment, which might not be very objective. Electroencephalography (EEG) provides a non-invasive method to measure neural activity with high temporal resolution. Spectral measures in the form of theta bands (4 – 8 Hz) have been suggested as a possible biomarker. This study focuses on classification of children with ADHD and those without ADHD using single band EEG feature. EEG recording was acquired from an open-access dataset. For each subject, absolute theta power for each electrode was calculated and analyzed using machine learning techniques through Neucom environment.**

Keywords*— Electroencephalography (EEG), Attention-Deficit/ Hyperactive disorder (ADHD), Theta band, Absolute power, Classification, Machine Learning, Neucom

I. Introduction and Rational

Attention Deficit / Hyperactivity Disorder (ADHD) is one of the most common neurodevelopmental conditions affecting children. It is marked by symptoms of inattention, hyperactivity, and impulsivity [1]. It often continues into adolescence and adulthood. Diagnosing ADHD is still difficult because it mainly depends on behavioural assessment. The methods are subjective and can differ from one practitioner to another. As a result, there is a high demand for neuropsychological markers that can help diagnose and monitor ADHD.

Electroencephalography (EEG) has become a useful tool for measuring brain activity directly and non-invasively. It offers excellent temporal resolution. Previous studies have shown that ADHD is linked to changes in spectral patterns, particularly with higher theta band activity in frontal and central regions [2]. This gives us a reason to explore theta power as a potential feature for classifying ADHD and control individuals.

The progress in machine learning provides a powerful tool for analyzing EEG data, moving past traditional approaches. By using algorithms like support vector machines, multilayer perceptron, we can identify complex and nonlinear relationship in EEG features that simple approaches might miss. Applying this algorithm to theta band power can lead to a thorough classification of ADHD and control individuals.

The chosen topic is highly relevant as it brings together neuroinformatics, which involves EEG data processing and Artificial Intelligence through machine learning techniques. This combination offers insight into ADHD-related brain activity and practical assessment of tools for classification.

II. Literature Review

Electroencephalography (EEG) is a widely used method for examining brain functions in children who are diagnosed with Attention – Deficit/Hyperactivity Disorder (ADHD). Studies indicate that children with ADHD typically exhibit increased theta activity (4-8 Hz) in comparison to non-ADHD children. Theta activity was consistently higher ADHD group, implying that it may serve as a possible marker to understand more about the disorder [2].

A related question in EEG research is whether to use absolute power (the direct measurement of signal strength in a frequency band) or relative power (power of frequency band divided by the total power across all bands). When comparing absolute and theta power, it was observed that in children, absolute theta power, and Theta-Beta Ratio (TBR) were significantly higher in ADHD group, while relative power showed no significant difference [3]. This suggests that normalizing data has its own advantages, absolute theta power can more effectively show the difference between groups, particularly in children.

Machine learning techniques are being used more to analyze EEG. They help to classify the differences between ADHD and control groups. Traditional classifiers like support vector machine (SVM), K-nearest neighbors (kNN), and multilayer perceptron (MLP) have been trained on EEG features, including theta power and theta/beta ratios. For instance, XGBoost was used on 19-channel EEG band features and reached 90.81% accuracy with leave-one-subject-out cross validation. SHapley addictive exPlanation (SHAP) was also performed to find important contributing features [4].

Taken together, these studies show that EEG band features are valuable for classifying ADHD and control groups. Machine Learning techniques are especially effective with feature extraction.

III. Spatio-Temporal Brain Data

A. Data Collection

The EEG data used in this study was obtained from IEEE DataPort open-access repository called “EEG data for ADHD/Control children” [5]. The dataset includes EEG recordings from children with clinically diagnosed Attention

Deficit / Hyperactivity Disorder (ADHD) and from healthy control individuals. Researchers recorded the signal using a 19-channel EEG cap, which was positioned according to the international 10-20 standard as shown in Figure 1. It is a standard method in neurophysiological research. The signal was sampled at 128 Hz. The data files are in ".mat" format, with each file representing one subject [5].

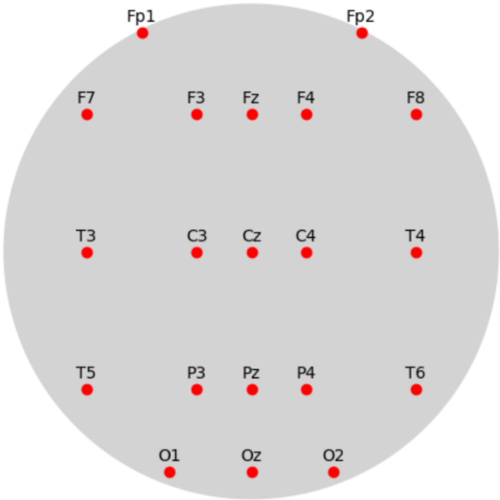

Figure 1: Electrode locations of the International 10–20 system for EEG.

The EEG recordings were based on visual attention tasks because visual attention is one of the deficiencies in children with ADHD. The children were asked to count the number of cartoon characters in a collection of photographs that were displayed to them. The images were large enough for children to easily see and count the characters, and the number of characters in each image was chosen at random from 5-16. Each image was shown continuously to provide continuous simulation during the signal recording [5].

B. Data Variable

The primary variable in the data set are the EEG voltage signals recorded from each electrode. These signals are measured in microvolts (μV).

Spatio Components: The spatio aspects of the data come from the 19 electrode channels. Each electrode corresponds to different brain region Fz, Cz, Pz, C3, T3, C4, T4, Fp1, Fp2, F3, F4, F7, F8, P3, P4, T5, T6, O1, O2. These channels show the frontal, central, parietal, and occipital regions. The standard 10-20 standard electrode placement was used [5].

Temporal Components: The EEG data is a continuous time-series signal with a sampling rate of 128 Hz [5]. This enables tracking of brain activities at milliseconds level. For this experiment, the focus was on the theta frequency band (4-8 Hz), which was extracted from the EEG signal. Both relative and absolute theta power can be extracted, but absolute power was used.

C. Previous Research on the Dataset

Several studies have used the same EEG dataset from IEEE DataPort that this project uses.

A recent machine learning framework was applied to the same dataset. They used Random Forest (RF), XGBoost, CatBoost, and LightGBM for classification and SHAP algorithm was used to assess the contributing features. CatBoost received the highest classification accuracy of 99.58% [6].

Min Feng and Juncai Xu transformed the raw EEG into input for a ConvMixer-based deep network, along with ECA attention. They trained and tested the model on the public IEEE DataPort ADHD dataset, which included 61 ADHD subjects and 60 controls, using 19 channels at 128 Hz. They reported strong classification results with spectrogram-based deep features. This paper shows a modern CNN-style method that directly applies to the same recordings [7].

A study used an autoencoder to extract features and a modified ResNet with double-augmented attention to classify ADHD and control groups using the same dataset. Their model achieved very high performance, showing the effectiveness of attention-augmented deep networks [8].

IV. DATA ANALYSIS

A. Study Design

The main goal of this study was to determine if absolute theta band power from EEG data could be a reliable feature for classifying ADHD and control subjects through machine learning methods. The dataset included 19 absolute theta power features for each subject. These features corresponded to the standard 10-20 EEG channels as shown in Figure 1. Each subject was represented as a single feature vector, with the group label (ADHD or Control) shown in the last column. All subjects were included without selecting features to keep the full spatial distribution.

The study was set up as follows:

1. Input features: Absolute theta power values from all 19 EEG channels.
2. Output target: Categorical class label (1 = ADHD, 2 = Control).
3. Tools: All experiments took place in the NeuCom environment. This environment provides access to traditional machine learning models and evolving connectionist systems.
4. Cross Validation: Leave-One-Out Cross-Validation (LOOCV) method was used. This ensured that each subject served once as a test case, while all other subjects were used to train the model.

B. Data Preprocessing

The EEG recordings came in ".mat" format. Each file represented one subject, either ADHD or Control. The spatial aspect of the data comes from the 19 electrode channels. For each channel, we calculated the absolute theta power (4-8 Hz). To stabilize the scale, we log-transformed all power values (log10). The final dataset was exported as a .csv file where each row represented one subject, the 19 log-transformed theta features were the inputs, and the class label (ADHD=1 or Control=2) was the output variable. This was necessary for data to be loaded into Neucom software [9].

C. Data Visualisation

Before classification, the dataset was examined with basic visualization methods to gain a clearer understanding of the distribution of subjects and features.

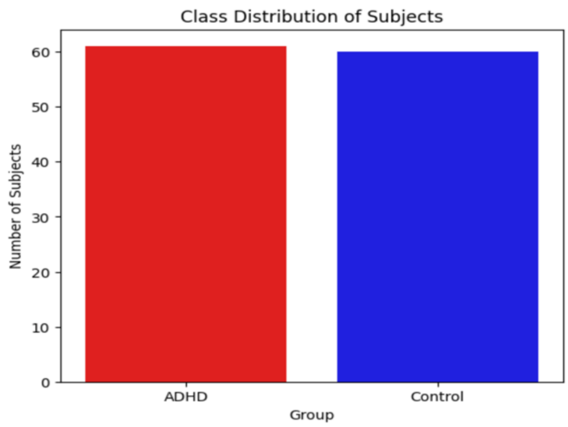

Figure 2: Class Distribution of ADHD and Control Subjects

The class distribution in Figure 2 displays the number of ADHD and control participants. The dataset is balanced between the two groups, which helps prevent the machine learning models from favoring one class. A balanced dataset matters because it lets the accuracy show true discrimination ability, instead of just predicting the majority class.

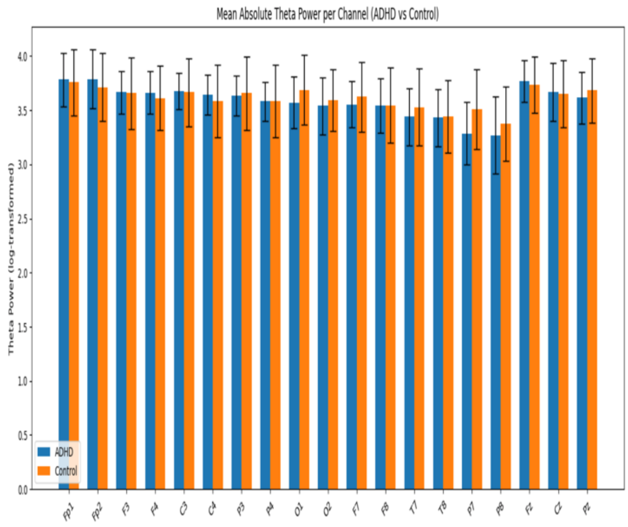

Figure 3: Bar Plot of Mean Absolute Theta Power across 19 EEG Channels

The grouped bar plot in Figure 3 compares the mean log-transformed absolute theta power across the 19 EEG channels for ADHD and control subjects. Each channel has two bars (ADHD and Control), with error bars indicating the standard deviation. From the figure, we can see that the ADHD group shows slightly higher theta power than the control group, especially at the frontal (Fp1, Fp2, F3, F4) and central (C3, Cz, Fz, C4) electrodes. This aligns with earlier studies that connect ADHD with increased frontal theta activity [1], [2]. However, the differences are not significant, and the overlapping error bars indicate variability among individuals. This overlap suggests that while theta power provides some useful information, it is not enough for complete separation between ADHD and control subjects on its own.

These visualizations offered a clear view of the dataset. The class distribution confirmed its eligibility for classification analysis. The grouped bar plot showed significant differences between the ADHD and control groups. These differences justify using theta power features for machine learning-based classification.

D. Machine Learning Approaches

Several machine learning classifiers were applied in NeuCom [9] to the theta band features, using both linear and non-linear methods. The goal was not only to identify the best-performing model but also to compare how different strategies managed the dataset. The Table 1 show that the overall classification accuracy ranged from about 65% to 99% across the models.

1. Multilinear Regression: It used a simple linear mapping between inputs and class labels. It provided an accuracy of 74.38%.
2. Support Vector Machines (SVM): Both polynomial (degree = 1) and radial basis function (RBF) kernels were tested. The polynomial kernel acts as a linear classifier and achieves moderate accuracy of 70.24%. The RBF kernel, on the other hand, provided good accuracy of 80.16% for gamma = 1 and 87.60 for gamma=2. This shows how important it is to capture non-linear relationships.
3. Multilayer Perceptron (MLP): A simple neural network architecture was tested with one and two hidden units. The MLP with two hidden units (accuracy of 76.03%) performed better than the version with one hidden unit (accuracy of 65.28%). This indicated that small neural networks could capture some complexity in the data, but they were still limited compared to SVMs.
4. Evolving Classification Function and Evolving Clustering Method for Classification: NeuCom's adaptive models were also used. These methods change their structure during training, which helps them respond to data complexity. Both models reached very high accuracy values of 99% and 95% respectively. However, such near-perfect results should be viewed carefully. They might show overfitting.

Table 1: Result of Various Machine Learning Technique used

Model	Key Settings	Accuracy (%)
Multilinear Regression	Default	74.38
SVM (Polynomial)	Degree = 1	70.24
SVM (RBF)	Gamma γ = 1	80.16
SVM (RBF)	Gamma γ = 2	87.60
MLP	1 hidden unit	65.28
MLP	2 hidden units	76.03
ECF	Default	99.00
ECMC	Default	95.00

E. Cross Validation

All models were tested using Leave-One-Out Cross-Validation (LOOCV). In this approach, one subject was left out from the dataset at each iteration and used as the test sample, while the remaining subjects formed the training set. This process was repeated until every subject had been tested once.

Leave-One-Out Cross-Validation (LOOCV) was chosen for several reasons:

1. It maximizes training data in each try, which is critical for relatively small EEG datasets.
2. It produces nearly unbiased estimates of model performance since each subject is tested independently.
3. It avoids problem with random variability found in train-tests splits, ensuring consistent evaluation across models.

Table 2: Results of LOOCV

Model	Key Settings	Accuracy (%)	Class 1 (ADHD)	Class 2 (Control)
Multilinear Regression	Default	59.50	62.30	56.67
SVM (Poly)	Degree = 1	61.98	65.57	58.33
SVM (RBF)	$\gamma = 2$	62.81	65.57	60.00
MLP	2 Hidden Unit	54.55	72.13	36.67
ECF	Default	62.81	63.93	61.67
ECMC	Default	58.68	60.66	56.67

The results in Table 2 show that the overall classification accuracy ranged from about 54% to 63% across the models.

1. Linear models (Multilinear Regression) reached about 59.5% accuracy. This indicates that simple linear decision was not enough to capture the differences between ADHD and control theta power.
2. Support Vector Machine (SVM) with a polynomial kernel slightly increased performance to about 62%. The SVM with an RBF kernel (γ=2) achieved the highest performance of 62.8%. This suggests that non-linear separation offers a benefit.
3. Multi-layer perceptron (MLP) with 2 hidden units achieved 54.5% accuracy. It showed a strong class accuracy imbalance, with 72.13% accuracy for ADHD and 36.67% for Control.
4. Evolving Classification Function (ECF) achieved performance close to the best SVM, which was about 62.8%. Unlike MLP, Evolving Classification Function (ECF) and Evolving Clustering Method for Classification (ECMC) models kept more balanced per-class accuracies. This balance is crucial for fair discrimination between ADHD and control.

Overall, the accuracy did not exceed about 63%. These results reflect the difficulty of classifying ADHD using EEG with a single frequency band, specifically theta. The findings show that theta power alone offers useful, but not enough, information for discrimination

V. Summary

This study focused at using absolute theta band EEG features to classify ADHD and control subjects with machine learning techniques. After preprocessing the EEG data into 19-dimensional feature vectors, the dataset was analysed in NeuCom [9] using several models, including multilinear regression, Support Vector Machine, Multi-layer perceptron, Evolving Classification Function and Evolving Clustering Method for Classification.

Exploratory data visualization showed that ADHD subjects had slightly higher theta power in frontal and central regions, compared to control individuals. The classification results differed between models. Standalone evaluations gave high accuracies, reaching up to 99%. However, LOOCV showed realistic performance, with most models getting between 58% and 63% accuracy. The best LOOCV results came from the support vector machine (RBF kernel, $\gamma = 2$) and the Evolving Classification Function model, both achieving approximately 62.8%.

These findings show that theta band features give useful but limited information for ADHD classification. This indicates that theta power alone cannot create a strong classification. Future work could include additional frequency bands, such as alpha and beta, along with connectivity measures and larger, more diverse datasets to improve classification performance.

VI. Future Work

This study focused only on absolute theta power. A simple next step would be to include additional frequency bands like alpha (8–12 Hz), beta (13–30 Hz), and gamma (>30 Hz). Previous studies have suggested that the theta/beta ratio might be a better marker of ADHD than theta alone [2], [3]. Adding multiple features would give a bigger picture of the EEG signal and could increase classification accuracy.

Another improvement would be to test feature selection methods to find out which EEG channels contribute the most to the classification. This might reduce noise in the data and make the models more efficient.

Different machine learning approaches could be used. For example, ensemble approaches like Random Forests or boosting methods could be tested. Similarly, deep learning techniques could also be introduced if larger dataset is available.

Although the current dataset is a great starting point, it has some limitations. A larger dataset with more subjects that includes various ages, genders, and ADHD subtypes would make the results more reliable. Recording EEG both while at rest and during simple tasks could show clearer differences.

References

[1] American Psychiatric Association, "What is ADHD?," 6 2022. [Online]. Available: https://www.psychiatry.org/patients-families/adhd/what-is-adhd. [Accessed 20 8 2025].

[2] A. R. Clarke, R. J. Barry, R. McCarthy and M. Selikowitz, "EEG analysis in Attention-Deficit/Hyperactivity Disorder: a comparative study of two subtypes," *Psychiatry Research,* vol. 81, no. 1, pp. 19-29, 1998.

[3] S. Markovska-Simoska and N. Pop-Jordanova, "Quantitative EEG in Children and Adults With Attention Deficit Hyperactivity Disorder: Comparison of Absolute and Relative Power Spectra and Theta/Beta Ratio," *Clinical EEG and Neuroscience,* vol. 48, no. 1, pp. 20-32, 11 5 2016.

[4] J. W. Kim, B.-N. Kim, J. I. Kim, C.-M. Yang and J. Kwon, "Electroencephalogram (EEG) Based Prediction of Attention Deficit Hyperactivity Disorder (ADHD) Using Machine Learning," *Neuropsychiatric Disease and Treatment,* vol. 21, pp. 271-279, 31 12 2025.

[5] A. M. Nasrabadi, A. Allahverdy, M. Samavati and M. R. Mohammadi, "EEG data for ADHD / Control children," 2020. [Online]. Available: https://dx.doi.org/10.21227/rzfh-zn36. [Accessed 03 08 2025].

[6] Y. Mao, X. Qi, L. He, S. Wang, Z. Wang and F. Wang, "Advanced machine learning techniques reveal multidimensional EEG abnormalities in children with ADHD: a framework for automatic diagnosis," *Frontiers in Psychiatry,* vol. 16, 2025.

[7] M. Feng and J. Xu, "Electroencephalogram-Based ConvMixer Architecture for Recognizing Attention Deficit Hyperactivity Disorder in Children," *Brain Sciences,* vol. 14, no. 5, 7 05 2024.

[8] J. Bansal, G. Gangwar, M. Aljaidi, A. Alkoradees and G. Singh, "EEG-Based ADHD Classification Using Autoencoder Feature Extraction and ResNet with Double Augmented Attention Mechanism," *Brain Sciences,* vol. 15, no. 1, 2025.

[9] KEDRI, Auckland University of Technology, "NeuCom – A Neuro-computing Decision Support Environment," [Online]. Available: https://kedri.aut.ac.nz/research-groups/data-mining-and-big-data-group/neucom-a-neuro-computing-decision-support-enviroment.